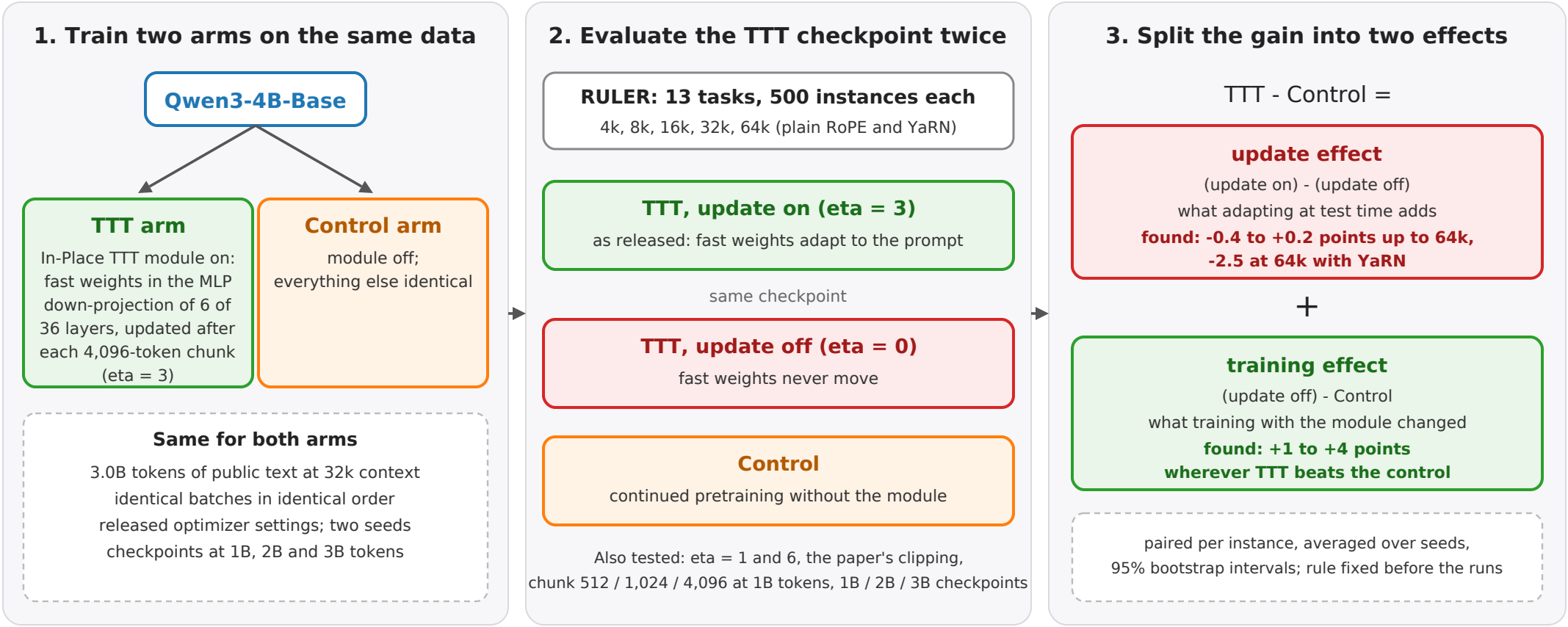
1. Train two arms on the same data
Qwen3-4B-Base
TTT arm
In-Place TTT module on: fast weights in the MLP down-projection of 6 of 36 layers, updated after each 4,096-token chunk (eta = 3)
Control arm
module off; everything else identical
Same for both arms
3.0B tokens of public text at 32k context
identical batches in identical order
released optimizer settings; two seeds
checkpoints at 1B, 2B and 3B tokens
2. Evaluate the TTT checkpoint twice
RULER: 13 tasks, 500 instances each
4k, 8k, 16k, 32k, 64k (plain RoPE and YaRN)
TTT, update on (eta = 3)
as released: fast weights adapt to the prompt
same checkpoint
TTT, update off (eta = 0)
fast weights never move
Control
continued pretraining without the module
Also tested: eta = 1 and 6, the paper's clipping,
chunk 512 / 1,024 / 4,096 at 1B tokens, 1B / 2B / 3B checkpoints
3. Split the gain into two effects
TTT - Control =
update effect
(update on) - (update off)
what adapting at test time adds
found: -0.4 to +0.2 points up to 64k,
-2.5 at 64k with YaRN
+
training effect
(update off) - Control
what training with the module changed
found: +1 to +4 points
wherever TTT beats the control
paired per instance, averaged over seeds,
95% bootstrap intervals; rule fixed before the runs